

The background is an abstract, painterly landscape. It features rolling hills and valleys in various shades of green, from light lime green to dark forest green. The top of the image is a pale, hazy yellow, suggesting a bright sky or a misty atmosphere. The overall style is soft and atmospheric, with visible brushstrokes or blended colors.

Head-Count Rules for Evolutionary Rescue

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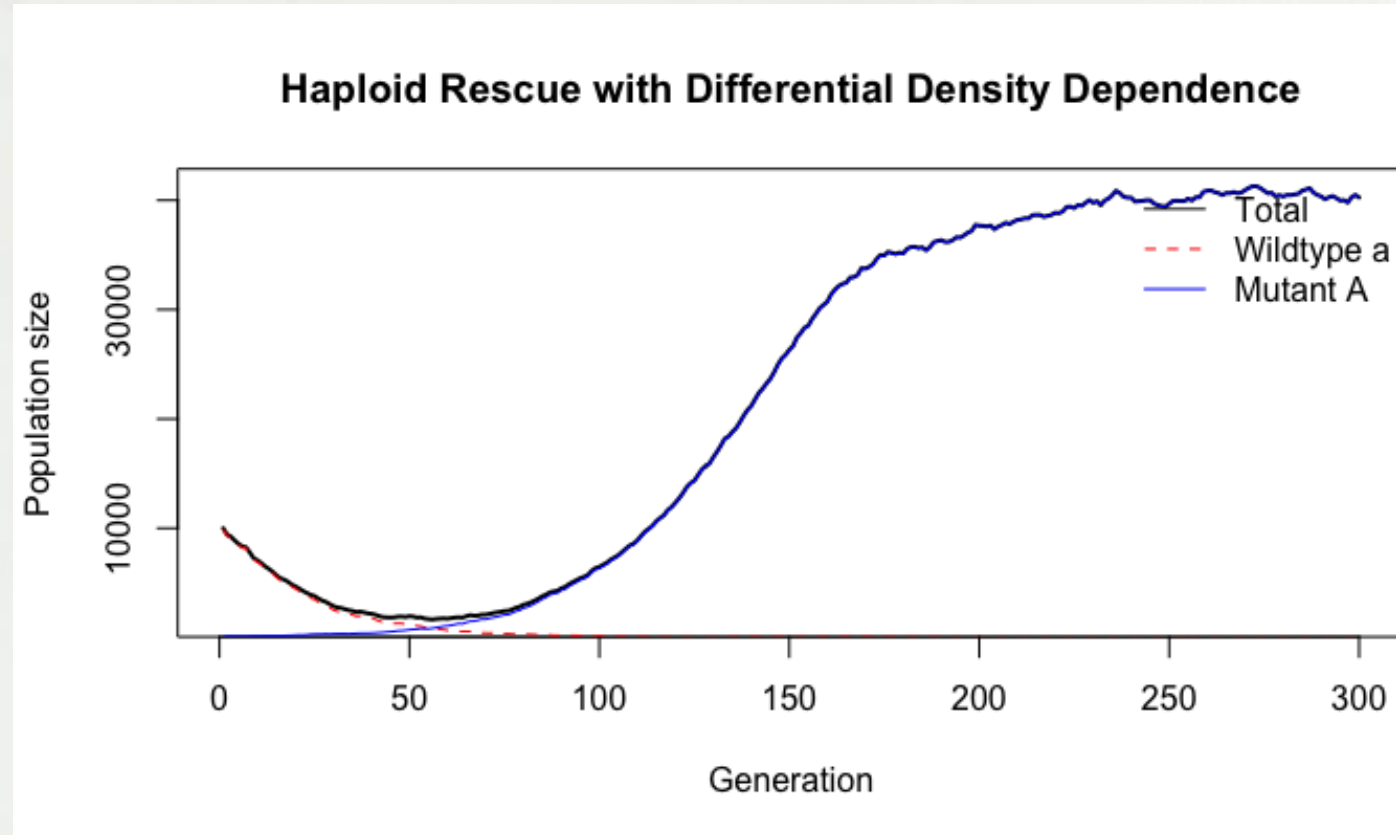


01

| Rescue in a Nutshell

Demographic Bottlenecks vs. Adaptation

Bottlenecks shrink the pool of beneficial alleles, making extinction more likely than adaptation. Microbial experiments show that **absolute copy number**, not percentage dilution, determines if resistance evolves. This prompts a critical question: what minimum head-count guarantees **$\geq 90\%$ evolutionary rescue** under density regulation?



Hard vs. Soft Rescue Modes



Hard Rescue

Relies on a **single mutant copy** sweeping to fixation.
Probability scales linearly with copy number, making it vulnerable to stochastic loss.



Soft Rescue

Requires **multiple copies** that collectively drive recovery, buffering against early stochastic loss and enabling faster rebound.

Density dependence can erode per-capita growth, altering the scaling between initial frequency and rescue probability.



02

| Model Design

Simulation Framework: Wright-Fisher with Logistic Fitness



1. Fitness



2. Offspring



3. Selection

Effective fitness follows a logistic form, capturing density-dependent contest competition:

$$w_{eff}(g, N_{tot}) = 1 + (w_g - 1) \times (1 - f_g \frac{N_{tot}}{K})$$

PARAMETERS

```
N0  <- 10000  # initial total population size
p0  <- 0.01    # initial fraction of mutants
K   <- 50000  # carrying capacity

w_a <- 0.95    # intrinsic fitness of wildtype (below 1)
w_A <- 1.05    # intrinsic fitness of mutant (above 1)

f_a <- 1.0     # density-sensitivity of wildtype
f_A <- 1.25    # density-sensitivity of mutant

tmax <- 300    # number of generations
```


Experimental Parameter Sweeps

Stage 1: Threshold Sweep

Find 90% rescue threshold at $N_0 = 10,000$ across $p_0 = 1/N_0$ to 0.2.

Stage 2: Scaling

Vary N_0 from 200 to 40,000 to quantify scaling exponent β .

Stage 3: Crowding Sensitivity

Fix $p_0 = 0.01$, vary mutant crowding factor f_A from 0.5 to 3.0.

Stage 4: Rebound Dynamics

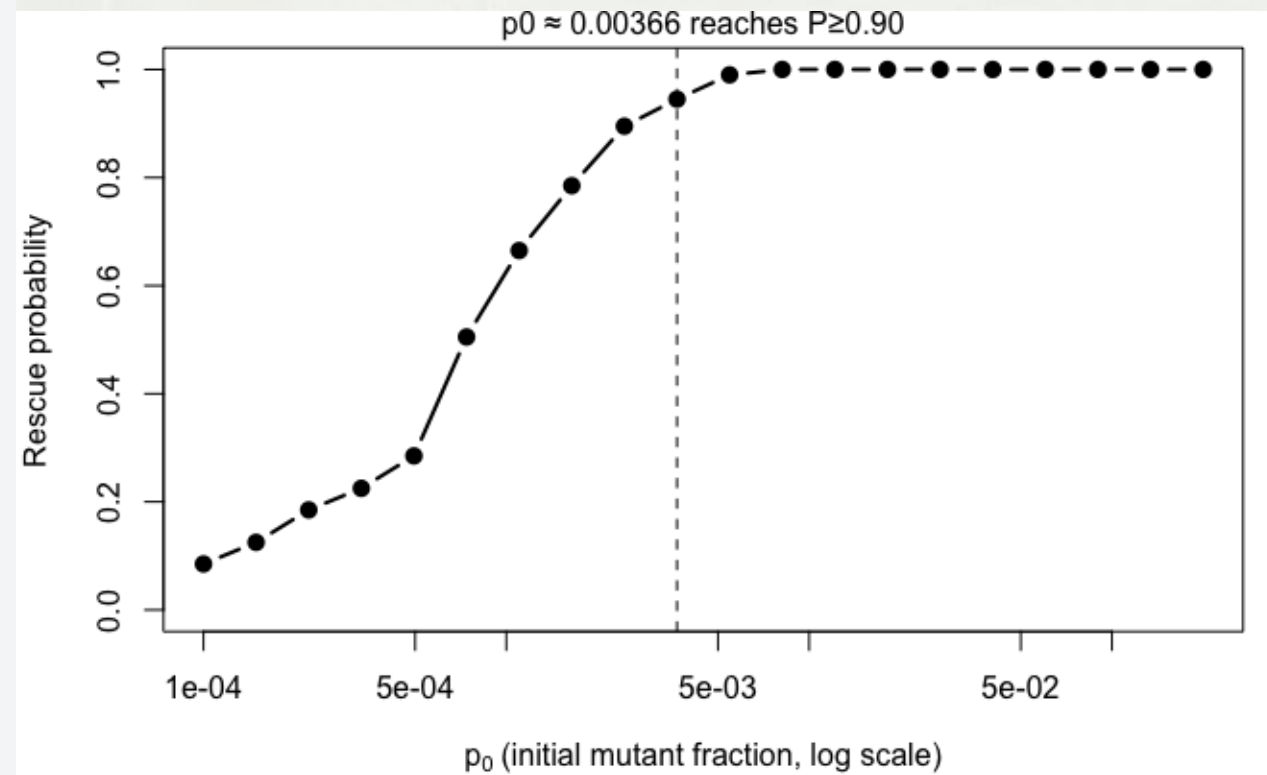
Track recovery time and minimum census for selected p_0 values.

200 replicates were run for each parameter combination.



03

| Key Results



Critical 90% Threshold

At $N_0 = 10,000$, rescue probability rises sigmoidally. The key breakpoint is:

$$p_0^* = 0.0037$$

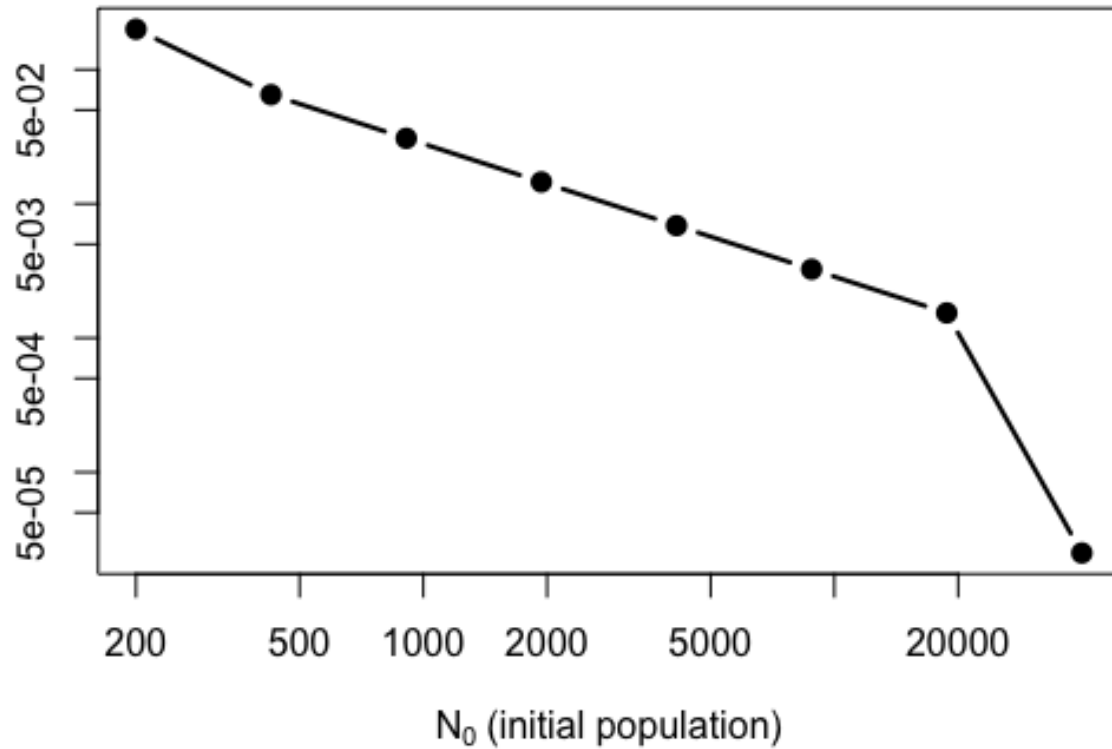
This corresponds to **37** mutant individuals.

This value falls within the predicted **20-60 head-count band**. Below $p_0 \approx 0.002$, drift usually erases mutants; above 0.01, rescue is almost certain.

Super-Inverse Scaling: A Universal Head-Count

The required fraction p_0^* scales with N_0 with a slope of **-1.30**, steeper than simple inverse (-1).

p_0 to reach $P[\text{rescue}] \geq 0.9$



Absolute Mutant Head-Count

Despite the super-inverse scaling, the absolute number of mutants remains constant.

Median: **25-30**

95% Prediction Band: 20-60

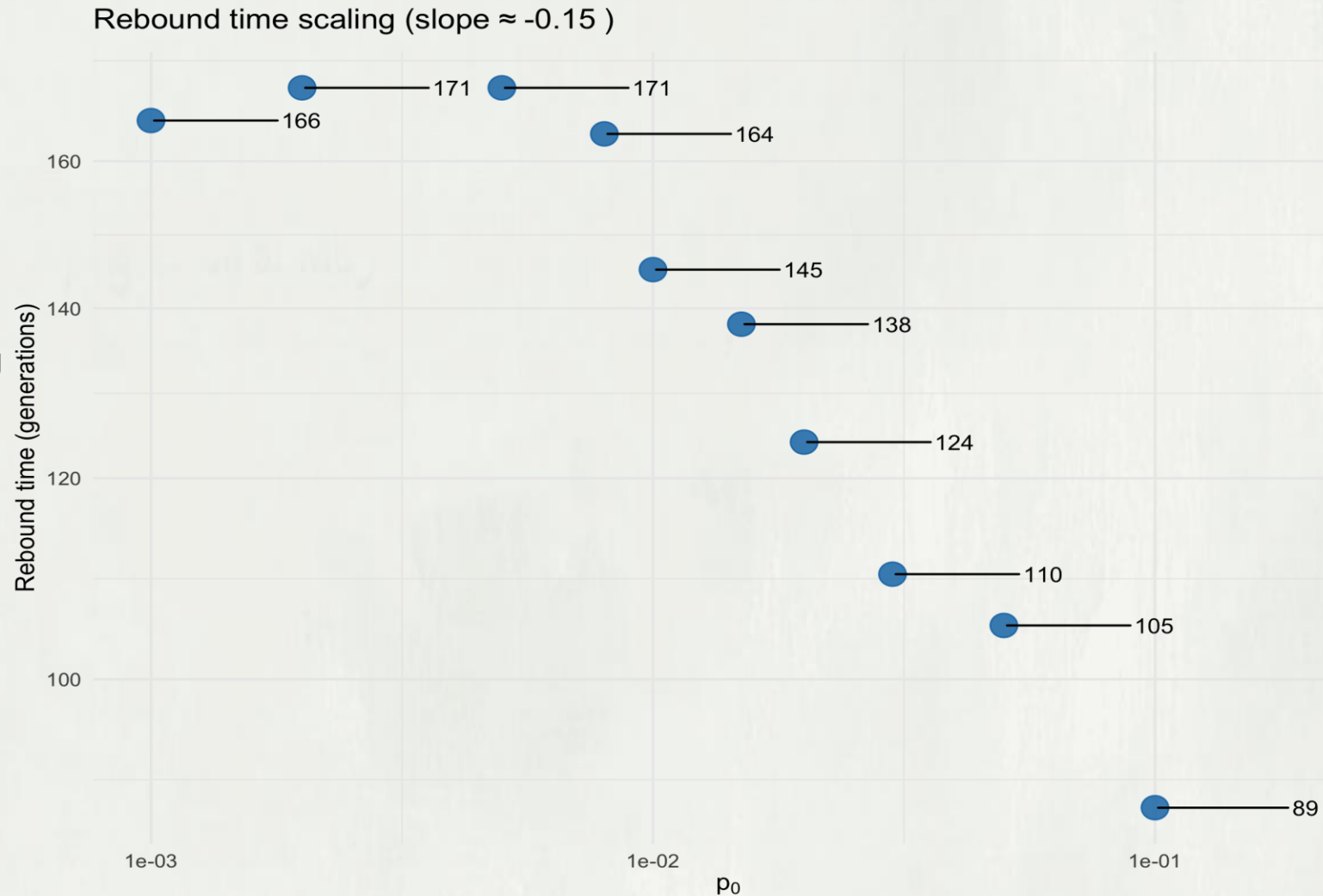
The Benefits of Higher Initial Frequency

Faster Recovery

Rebound time **halves** per log-unit increase in p_0 , dropping from ~160 to ~90 generations.

Demographic Insurance

Minimum census during decline rises from <100 to >1,000, buffering against stochastic loss.

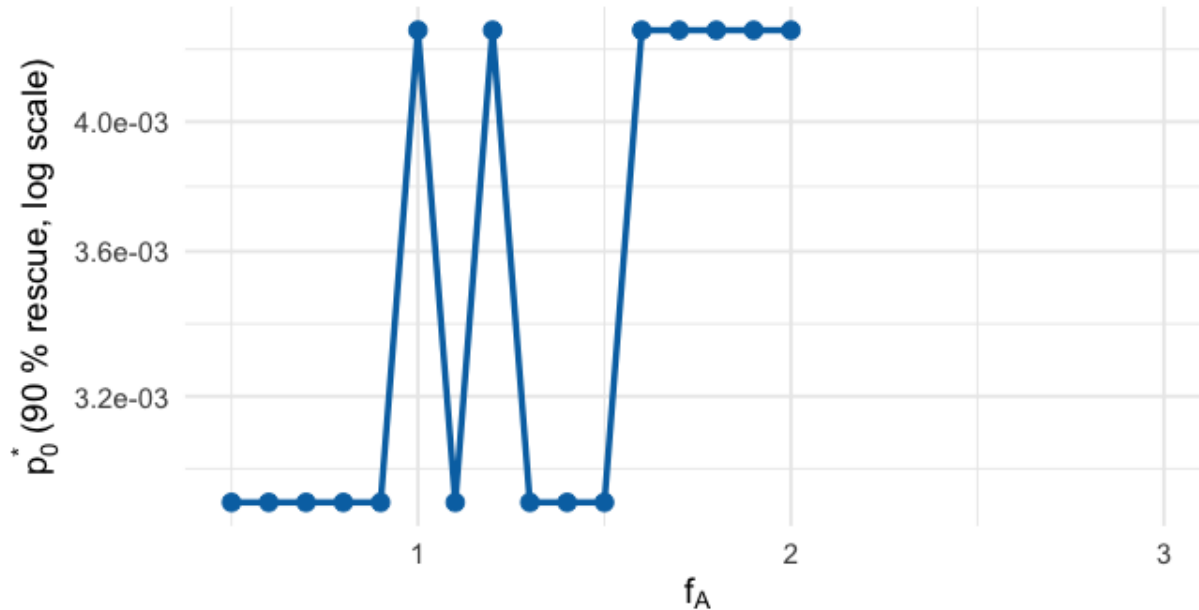


The threshold at $p_0 \approx 0.01$ marks the shift from uncertain recovery to rapid, reliable rescue.

Sensitivity to Mutant Crowding (f_A)

Minimal initial mutant fraction for 90 % rescue

p_0^* vs mutant density-sensitivity f_A



Stable Region ($f_A < 1.5$)

p_0^* is relatively constant. Demographic noise dominates.

Sensitive Region ($f_A > 1.8$)

Threshold rises steeply. Competitive suppression demands larger releases.

Key Insight

Underestimating f_A can **double** the required release size. Field calibration is crucial.



04



Implications

A Universal Head-Count Rule

The near-constant head-count (median ~30) is an emergent property of two opposing forces:

- ↓ **Weaker Demographic Noise:** In larger populations ($N_0 \uparrow$), drift effects are reduced ($\propto 1/\sqrt{N}$).
- ↓ **Stronger Density Feedback:** As $N/K \rightarrow 1$, mutant fitness is suppressed, offsetting the noise advantage.

This principle aligns with microbial studies, extending a universal $O(10-100)$ copy rule to vertebrate-like life histories.

Management Guideline for Genetic Rescue

Minimum Target

≥ 30

individuals for a $\geq 90\%$ rescue probability.

Conservative Target

≥ 60

when failure is irreversible or variance is high.

Never Use Percentage Targets

Releasing 0.1% of N_0 yields 2 mutants when $N_0=2,000$ (too few) but 40 when $N_0=40,000$ (excessive). **Always prioritize absolute head-count.**



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Next Steps

Caveats and Future Directions



Current Model

Minimalist by design: single locus, haploid, closed, non-spatial. This clarifies mechanism but has limitations.

Future Work Should:

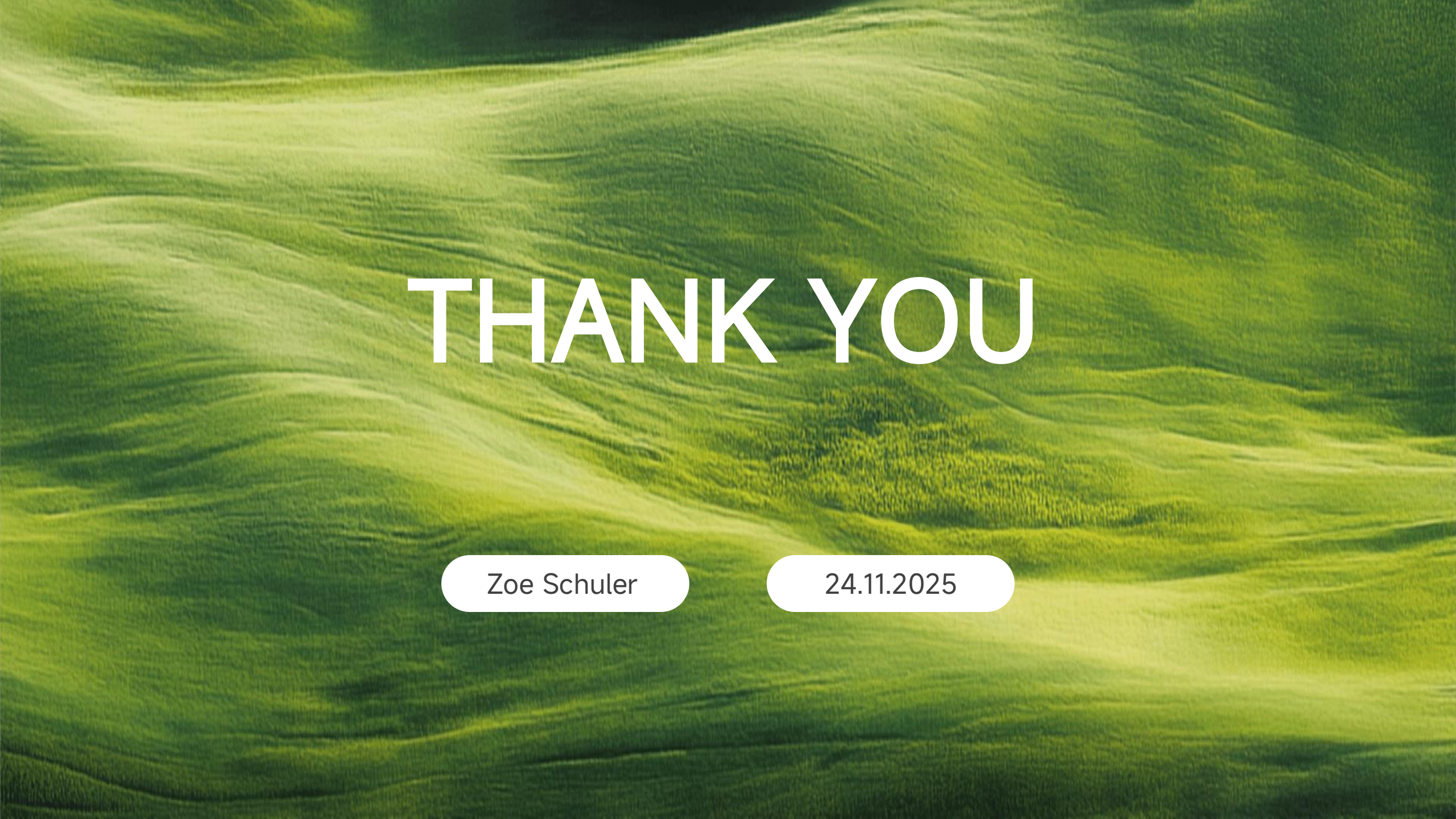
- ✓ Implement bootstrap resampling to quantify uncertainty.
- ✓ Use AIC to formally compare absolute vs. relative predictors.
- ✓ Test robustness to diploid genetics and recurrent mutation.
- ✓ Explore polygenic rescue and habitat heterogeneity.



The Head-Count Rule: A Simple Guideline

Evolutionary rescue probability is best predicted by **absolute copy number**, not frequency.

Across a wide range of scenarios, releasing **20-60** beneficial individuals provides a robust guideline for preventing extinction through genetic intervention.

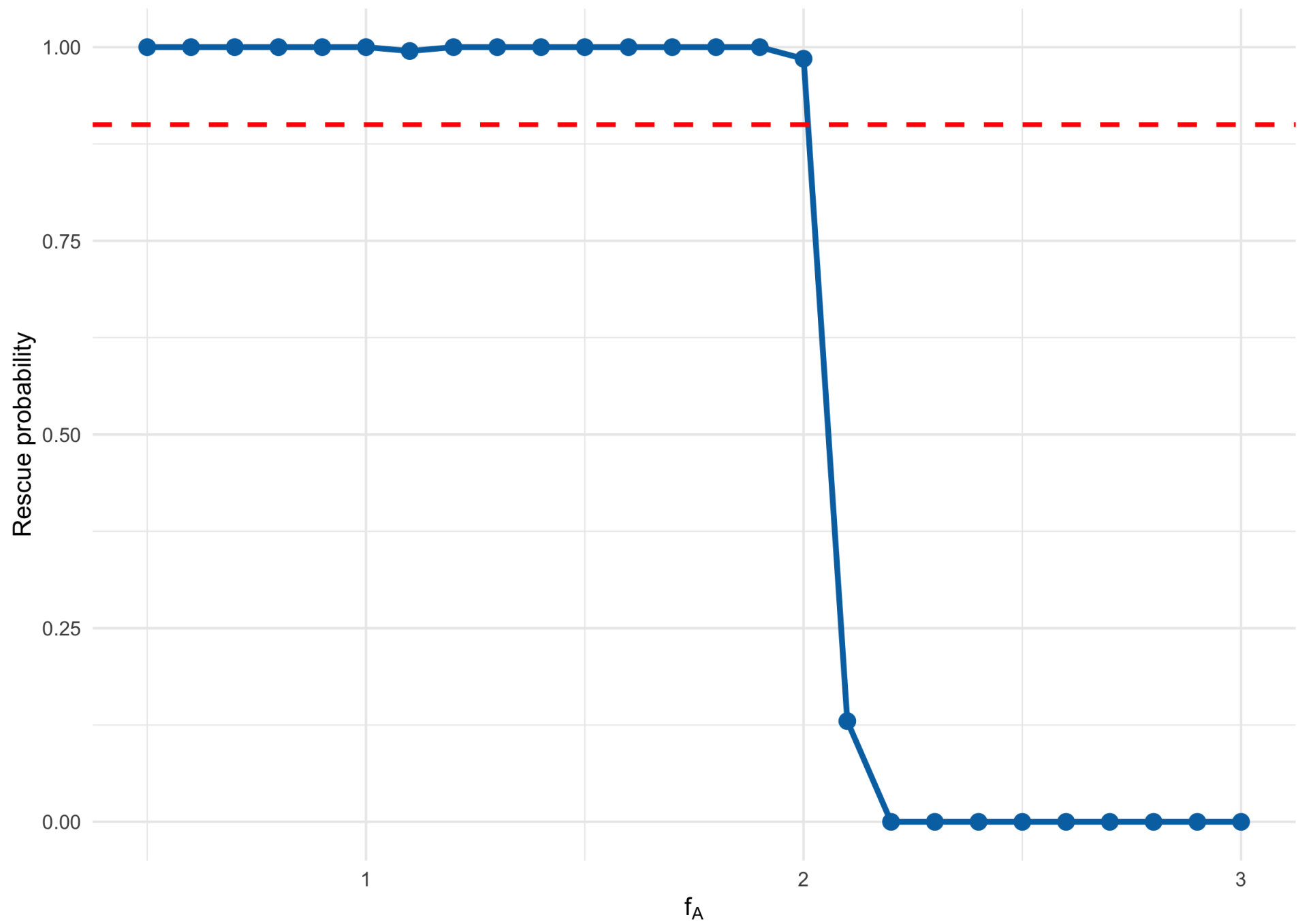
The background of the slide is an aerial photograph of rolling green hills, likely in a rural or agricultural area. The hills are covered in lush green grass, and the perspective creates a sense of depth and movement. The lighting is soft, highlighting the contours of the hills.

THANK YOU

Zoe Schuler

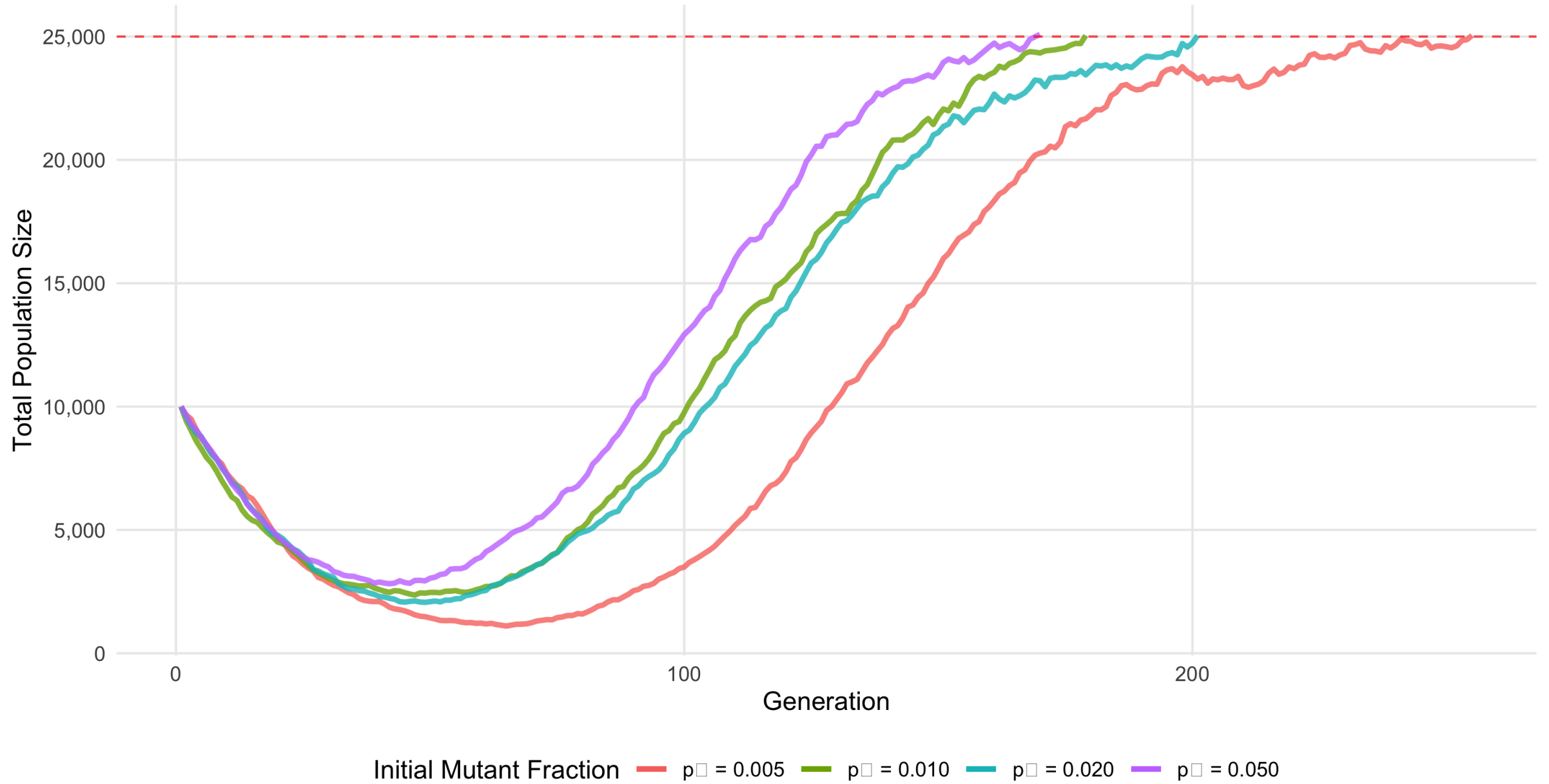
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Crowding sensitivity ($p_{\square}=0.01$)



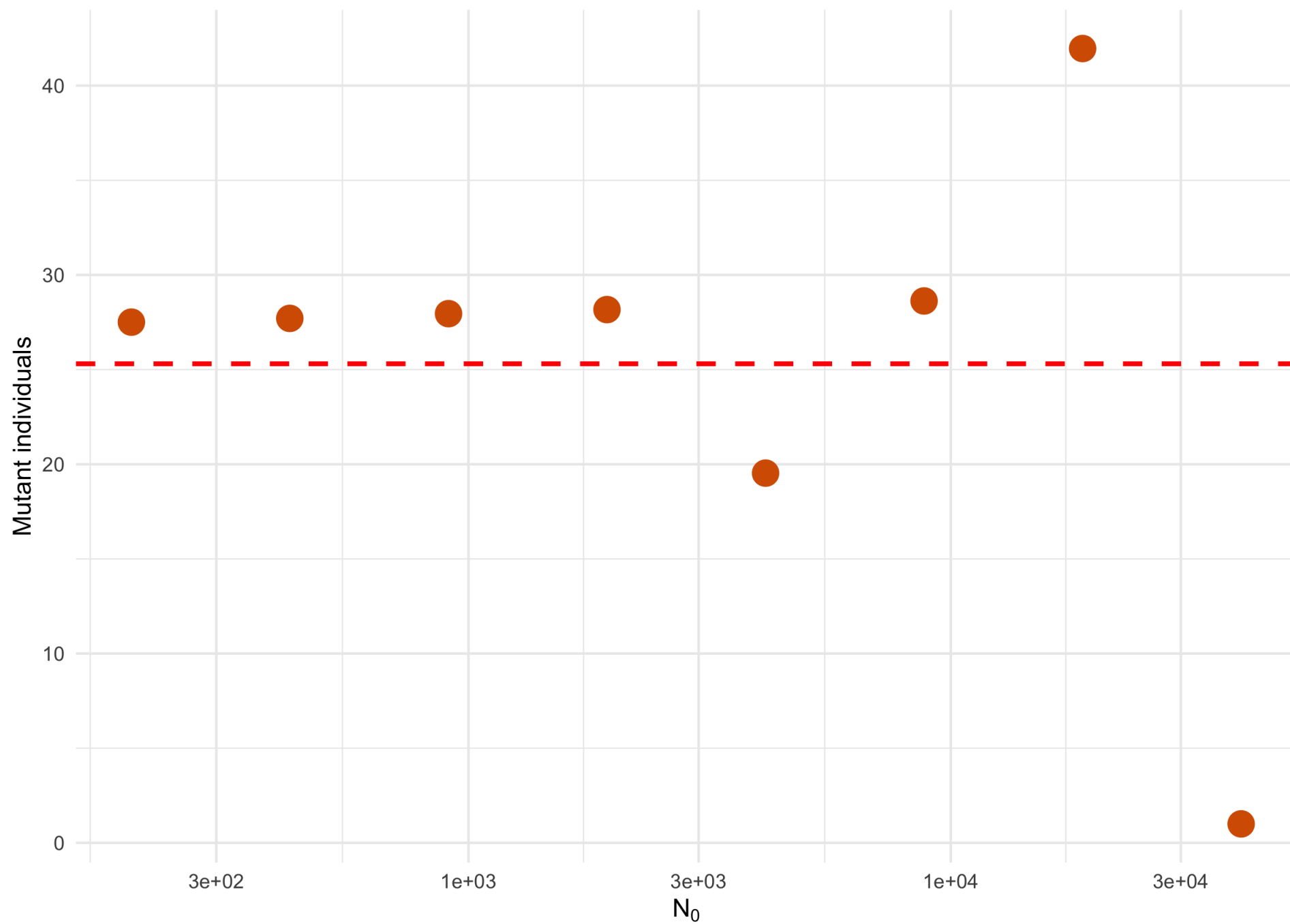
Population Dynamics: Single Population View

Total population trajectories showing rescue patterns

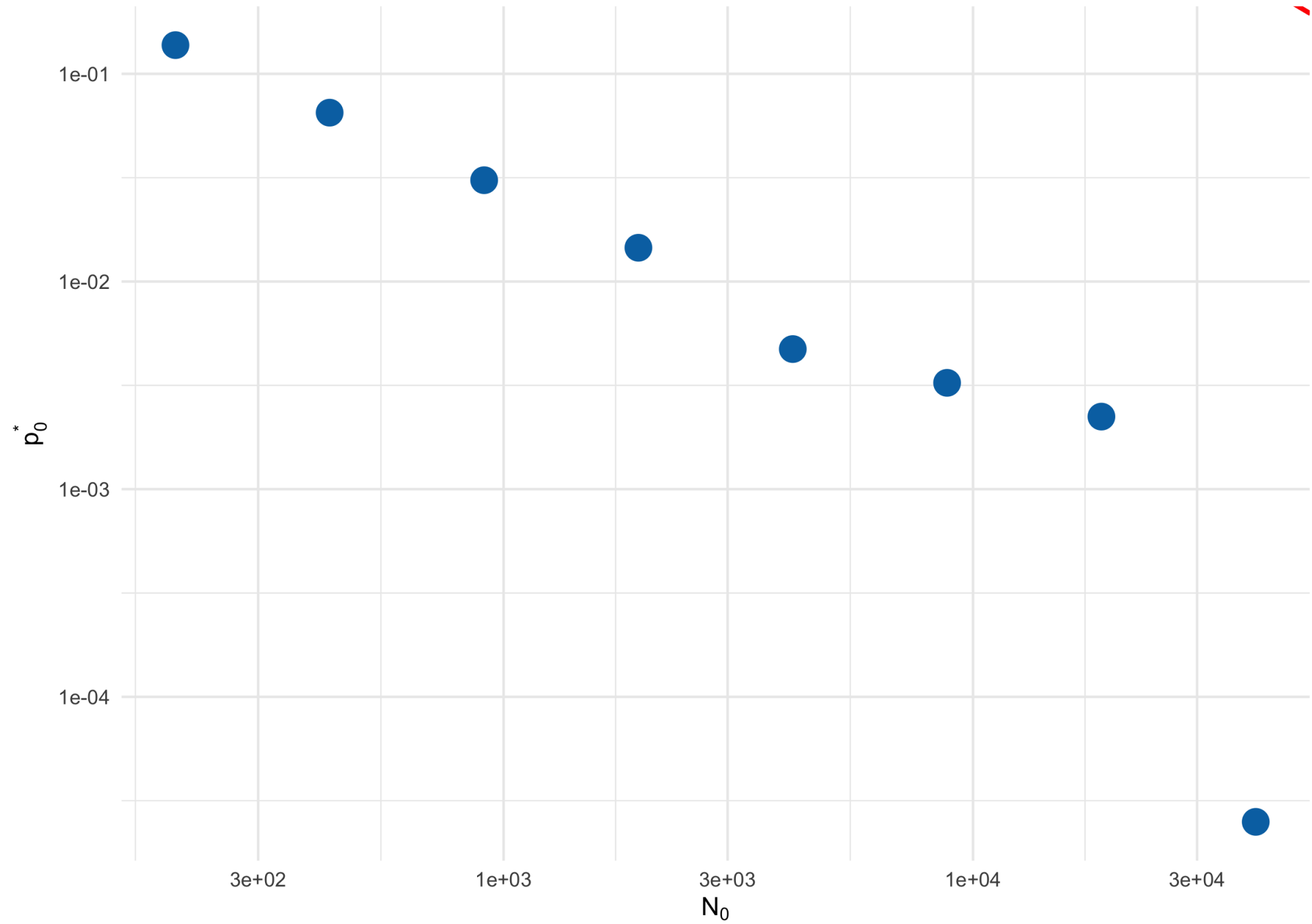


Red dashed line shows rescue threshold (50% of K = 50,000)

Constant headcount (median ~30)

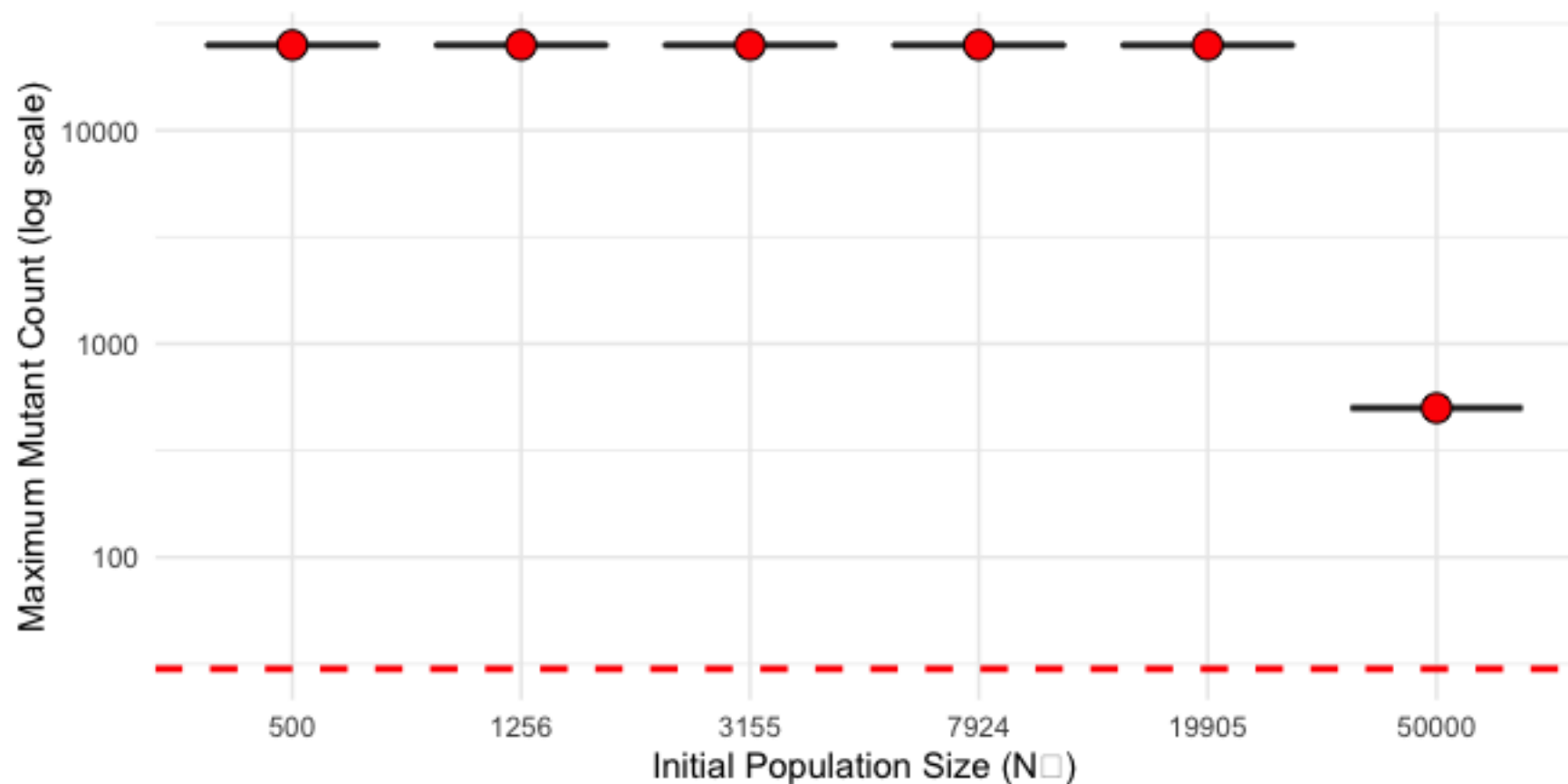


Super-inverse scaling (slope = -1.34)



Universal Head-Count Rule

Maximum mutant population size in successful rescues



Red dashed line at $N=30$ shows predicted median across all N_0 values